

Why and How to Compile a Custom Kernel

This article goes deeper into the Linux kernel and highlights the importance of building a custom kernel.



The Linux kernel was initially created and developed by Linus Torvalds in 1991. Today, the Linux kernel is ubiquitous; nearly 95 per cent of the supercomputers in the top 500 list and all Android phones use the Linux kernel. The Linux kernel (on Android) is also catching up with Microsoft Windows in the personal computing domains such as laptops and desktops after thrashing the Windows Mobile OS in the smartphones segment.

What is a kernel?

An operating system basically comprises two main software components—the kernel (core + system services) and the user interface (GUI or TUI). According to Wikipedia, the kernel is a computer program that manages input/output requests from software, and translates them into data processing instructions for the central processing unit and other electronic components of a computer. To put it in simple terms, the kernel is the core part of the OS, which abstracts the hardware. There are

basically three types of kernels: monolithic kernels, micro kernels and hybrid kernels. The basic difference between the monolithic and micro kernel is that in monolithic kernels, the entire OS runs in the kernel space, whereas in micro kernels, except for core services, most of the OS services run in the user space. There are advantages and disadvantages with both types of kernels. The Linux kernel belongs to the monolithic category. The proponents of micro kernel architecture believe that monolithic kernels are outdated. In a famous debate between Andrew S. Tanenbaum and Linus Torvalds, Tanenbaum argued that Linux was an outdated kernel even back in 1992 as it was a monolithic kernel. It is an interesting debate and I suggest you to search and read about this controversy.

So why should one care to configure, modify and compile Linux kernels? This question can be easily answered by looking at the variants of GNU/Linux distros available in the market. All these distros have their own